



The 3 March 2015 eruption of Villarrica volcano, Southern Andes of Chile: Overview of deposits and impacts



Romero, J.E.^{a*}, Keller, W.^b, Díaz-Alvarado, J.^a, Polacci, M.^c and Inostroza, M.^d

^aDepartamento de Geología, Universidad de Atacama, Copiapó, Chile
^bProyecto de Observación Volcán Villarrica (POVI), Pucón, Chile
^cSchool of Earth, Atmospheric and Environmental Sciences, University of Manchester, M13 9PL Oxford Road, Manchester, United Kingdom.
^dPrograma de Doctorado, Universidad Católica del Norte, Antofagasta, Chile.
* Contact e-mail: jorge.romerom@alumnos.uda.cl

ABSTRACT: Villarrica is the most active volcano of the Southern Andes Volcanic Zone and has sourced ~130 small eruptions between 1384 and 1984. After the deepening of its active lava lake in April 2012, an increase in the intensity of ASTER thermal anomalies in November and December 2014 was observed. Furthermore, a notable increase in the volcano activity occurred in February 2015, associated with the new rising of the lava lake, which resulted in Strombolian explosions. The paroxysmal phase occurred on 3 March 2015, and consisted of a short-lived (~18 min.) Strombolian-style activity culminating with a lava fountaining 1.5 km height and a column of ca. 11-12 km above the crater level. Spatter flows were formed, producing ice-melting and consequent lahar floods descending in five ravines around the volcano. Two of these lahars transported about 1 x 10⁶ m³ of sediments. From preliminary studies on scoria fallout deposits, a volume varying between 4.4 and 7.3 x10⁻³ km³ of tephra is calculated. In distal areas (20 km downwind), ash mostly consists of scoria fragments, volcanic glass drops and filaments. The mineralogical assemblage of spatter samples consists of plagioclase (plg) and olivine (ol) phenocrysts, occasionally showing resorption textures, in a microcrystalline groundmass. The impacts of this eruption include damage to some minor private and public infrastructure, evacuation of more than 3000 people and a long- lasting touristic impact due to closing of the volcano accesses.

INTRODUCTION

Villarrica stratovolcano (39°25'S, 71°57'W; 2847 m high) has evolved since Middle Pleistocene to Holocene [1] being the most active volcano (Fig. 1a) of the Southern Andes Volcanic Zone [2, 3]. Its evolution has been controlled by the building and destruction of old volcanic edifices, caldera and ignimbrite-forming eruptions [4, 1, 5, 6, 7]. The current summit hosts a 200 m diameter crater, containing an active and fluctuating, deep (<100 to 150 m) lava lake (Fig. 1b) or an open, degassing, vent [8]. Villarrica has sourced almost 14 moderate explosive eruptions in the last 14 ky [4], and ~100 small eruptions between 1384 and 1971 [9]. Mostly effusive and slightly explosive, Hawaiian to Strombolian eruptions have occurred in historical time. Products collected on 2004 by [10] included both scoria and golden pumice, fresh glassy coarse-ash with phenocrysts of olivine (ol) and plagioclase (plg). The latter paroxysmal eruption at Villarrica on 2015 possess a potential to study dynamics and mechanisms of basaltic explosive eruptions, from both erupted materials and volume of deposits. In the current contribution, we describe briefly the chronology, erupted products, deposits and impacts of the 3 March 2015 eruption.

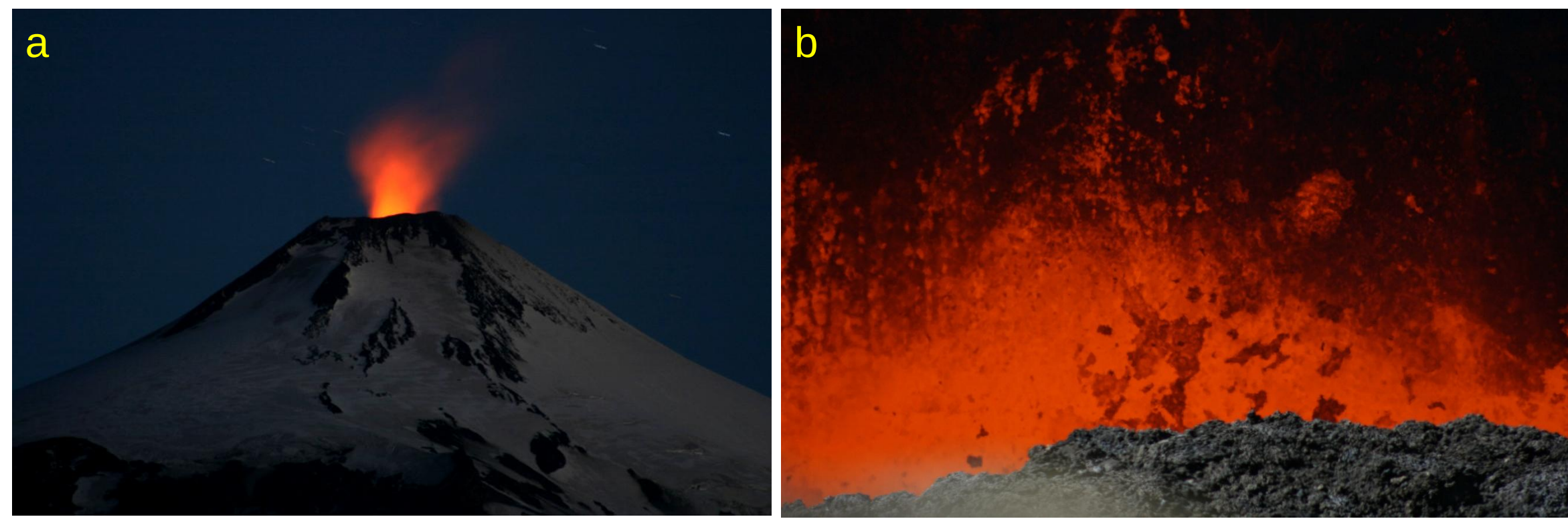


Fig. 1: Villarrica volcano. **a:** view of the active stratovolcano with incandescence at the summit. **b:** detail of the lava lake inside the crater. Photos by Patricio Arias.

THE 3 MARCH PAROXYSMAL 2015 ERUPTION

Prior to its 2015 unrest, Villarrica experienced low incandescence and deepening of its lava lake between April 2012 and the first days of 2015. Volcanic jetting was observed in December 2015 and seismic and surface activity increased dramatically from 1 to 16 February 2016 (Fig. 2a; <http://www.sernageomin.cl/volcanes.php>). Also, discrete Strombolian explosions occurred in February (Fig. 2b), and were associated to lava spatters emissions and MODIS satellite thermal alerts. Frequency and intensity of Strombolian explosions increased from 28 February up to 3 March, when a seismic swarm preceded a paroxysmal eruption [11] which started at 03.07 local time on 3 March (Fig. 2c) and lasted 18 min. A 1500 m high lava fountaining (Fig. 2c) feed by two closely-separated vents sourced this brief but violent Strombolian activity, with heavy tephra fall to the east and incandescent scoria flows predominantly on the north and east flanks [12]. The column height is calculated to be 6-8 km asl [11; 13] (Fig. 2d). Glacier melting produced lahars via the Voipir, Correntoso, Zanjón Seco-Carmelito, Pedregoso and Turbio rivers [11, 15]. Very low activity, characterized exclusively by weak ash plumes, was observed the following days (Fig. 2e).

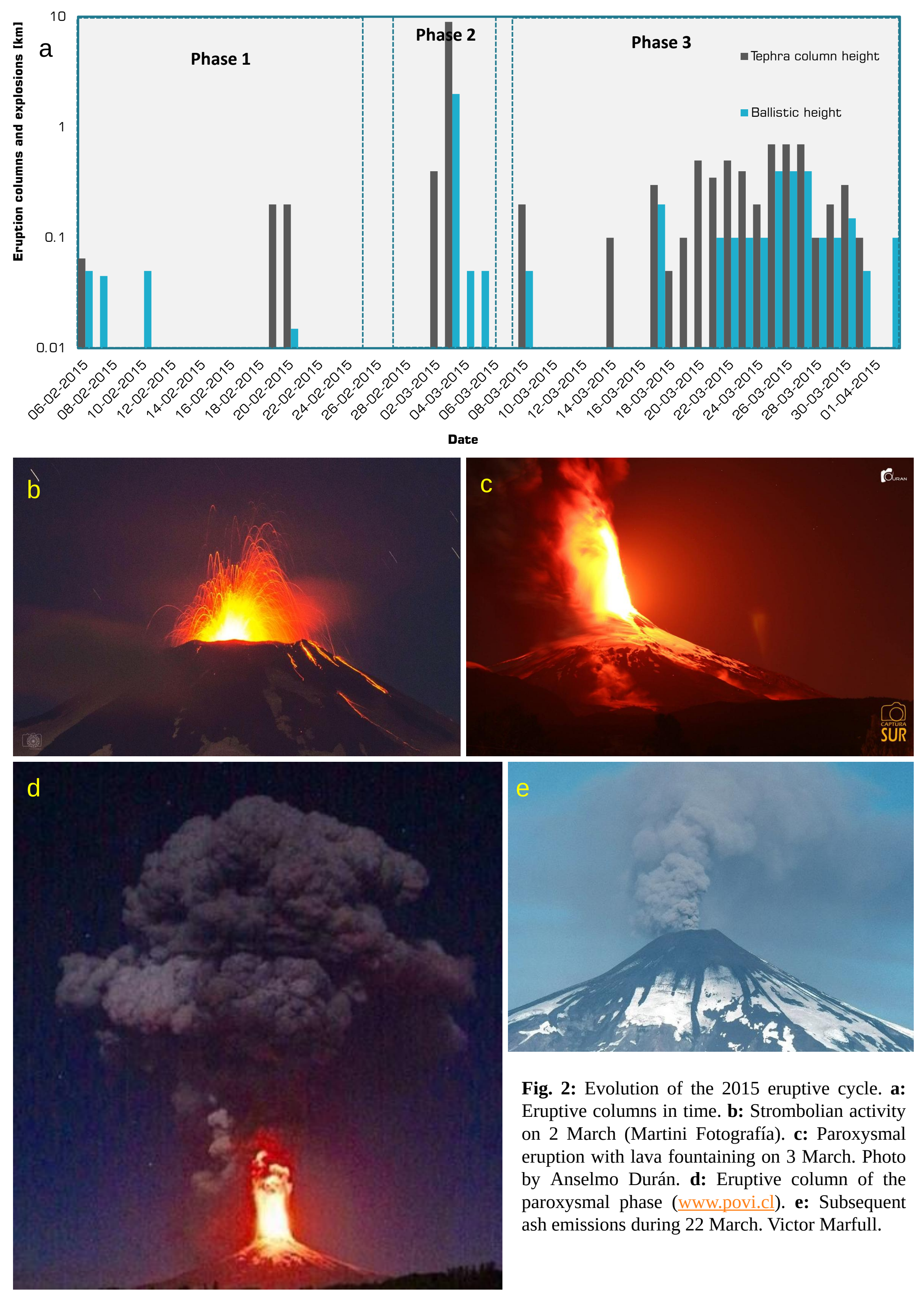


Fig. 2: Evolution of the 2015 eruptive cycle. **a:** Eruptive columns in time. **b:** Strombolian activity on 2 March (Martini Fotografía). **c:** Paroxysmal eruption with lava fountaining on 3 March. Photo by Anselmo Durán. **d:** Eruptive column of the paroxysmal phase (www.povi.cl). **e:** Subsequent ash emissions during 22 March. Victor Marfull.

PRODUCTS AND DEPOSITS OF THE ERUPTION

Lahars

According to the work of [15], the Zanjón Seco lahar (Fig. 3a1-2a3) has a volume ranging from 0.5 to 0.7 x 10⁶ m³, with velocities between 5 and 11 m/s and waves with 6 m height (Fig. 3a1). Pedregoso river lahar, was estimated in a volume of 0.3 to 0.4 x 10⁶ m³, with a velocity of 12 m/s [15]. At Turbio river bridge (Fig. 3b1), distant 18 km from the crater, the maximum wave height measured was 3 m and boulders of 1.5 m diameter were transported (Fig. 3b2). Upstream (6 km from the crater), the deposit thickness reached > 1 m and was compound by polymictic meter-sized blocks (Fig. 2b3).

Scoria fall deposits

Field observations carried out on March 10 and 11, in the downwind regime (Fig. 3c1-3c5), allows a basic hand-reconstruction of the five scoria fall isopachs between 0.1 and 5 cm (Fig. 3). Scoria forms a tephra fall deposit toward the southeast (Fig. 3c3). Distal ash is made of scoria fragments, irregular and broken volcanic glass drops (Fig. 4a), some of them elongated (Fig. 4b). A spatter sample (Fig. 3c6) thin section corresponds to a bomb-sized glassy crystalline tuff with a 20% of phenocrysts. From these, 13% are subhedral plg crystals (Fig. 5a) with polysynthetic twinning (0.4 to 2.0 mm in size), 5% of subhedral to anhedral ol phenocrysts (Fig. 5b; 0.2 to 0.4 mm in size), which are generally fractured and a 2% of cpx anhedral crystals (Fig. 5c; 0.1 to 0.2 mm in size). The groundmass represent the 80% and is made of 35% of pl subhedral microlites (0.1 to 0.2 mm) and 45% of glass (Fig. 5d). Minimum bulk tephra volume was calculated using various methods (eg. 16, 17, 18) (Fig. 3), varies between 4.4 and 7.3 x10⁻³ km³ (Table 1). The composition of scoria correspond to medium-K basaltic andesite [19].

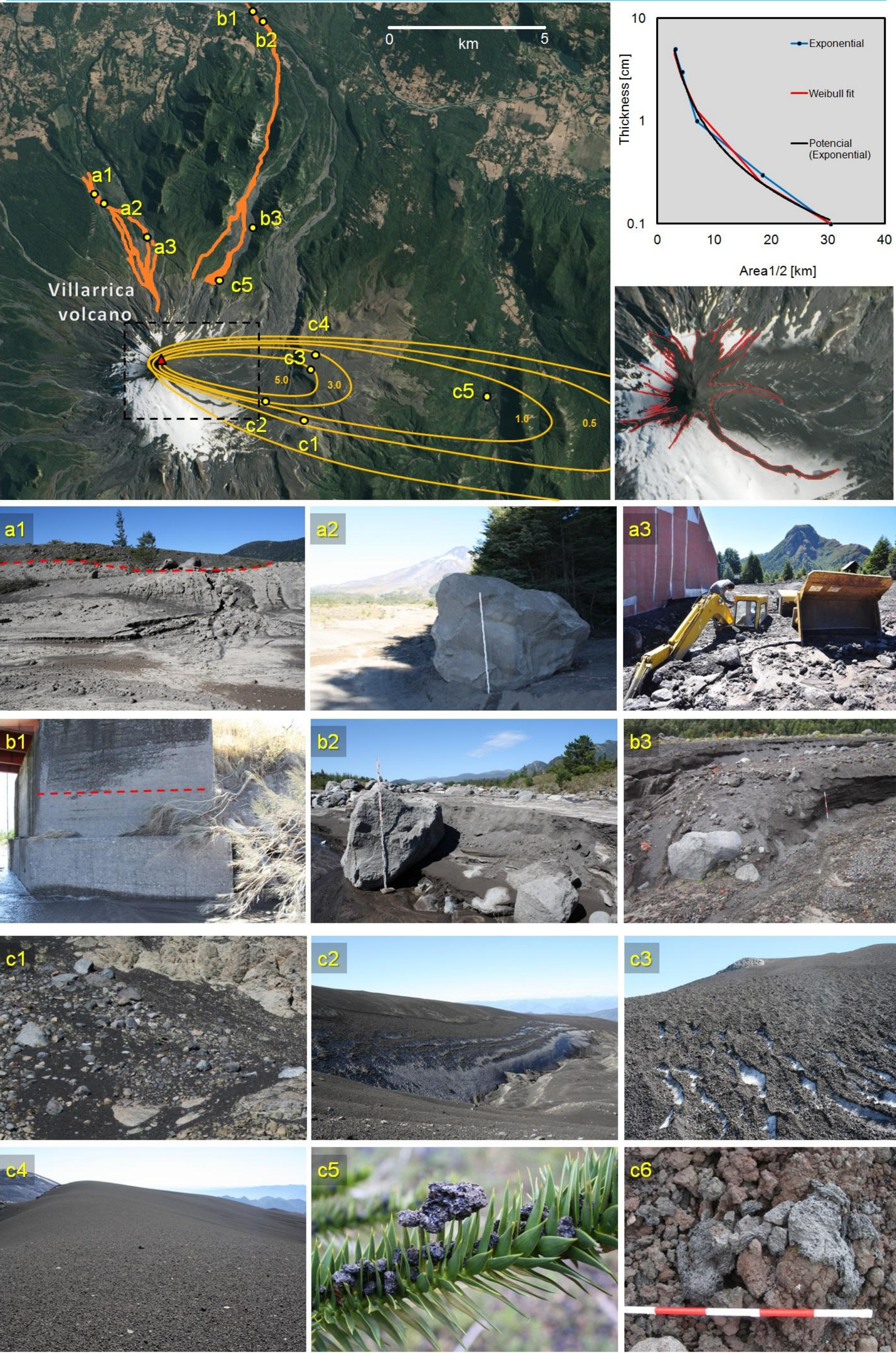


Fig. 3: Deposits of the 3 March 2015 eruption. Map above is after [15] **a:** Zanjón Seco Lahar. Photos by Werner Keller and Hugo Moreno (Courtesy of D. Basualto). **b:** Lahars of Pedregoso river. Photos by Werner Keller and Jorge Romero. **c:** Scoria fall deposits at southeast of the crater (see labels in map for location). Photos by Werner Keller.

Volcanic mixed avalanche

A detailed description of these deposits is in [19]. According to those authors, a series of radial deposits lengths of 0.7 to 3.5 km from the crater; Fig. 2e) consisting of scoria and ice/snow, with distinctive morphologies characterized by a thin (30-50 cm) pyroclastic coverage above the ice and snow, thick fronts (<3 m), levées and longitudinal ridges, were deposited during the climatic phase of the eruption.

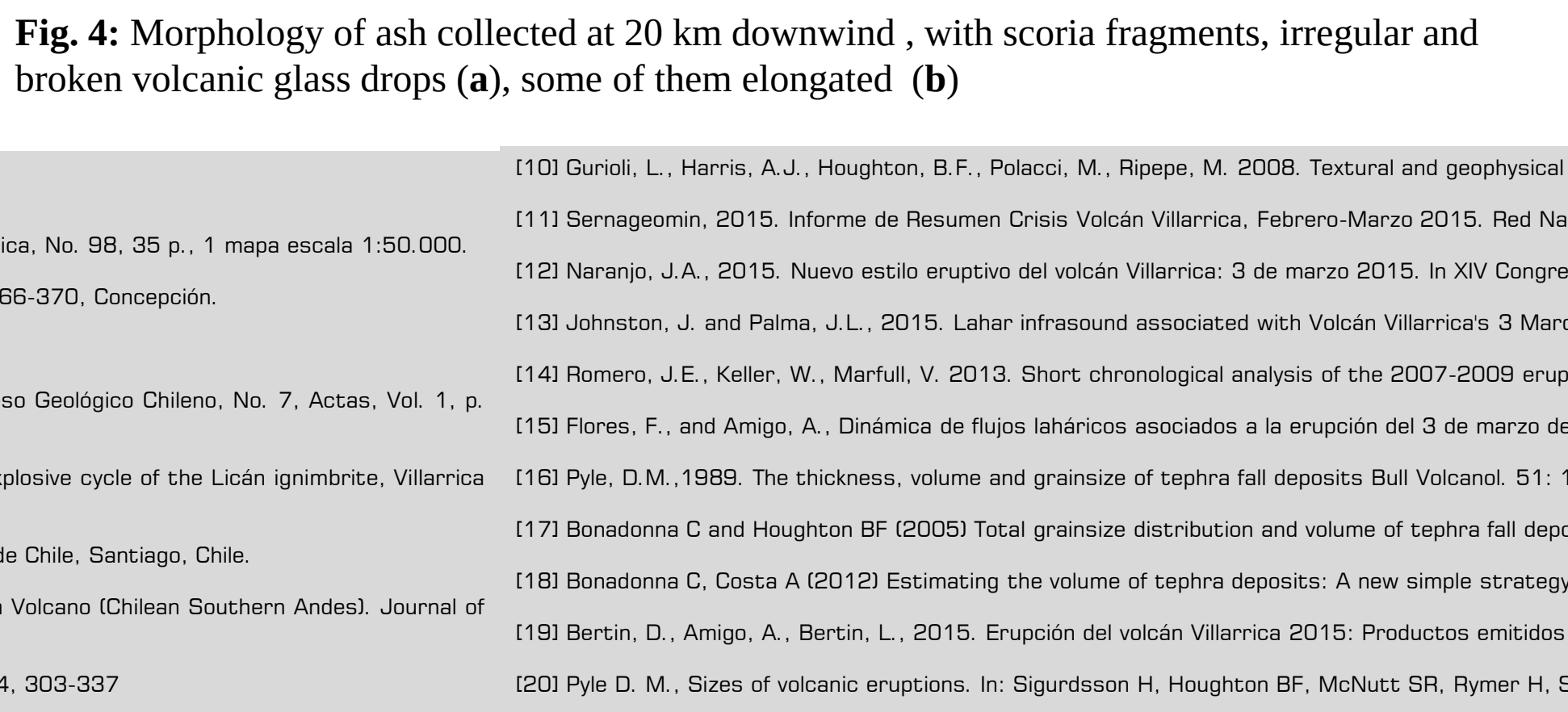


Fig. 4: Morphology of ash collected at 20 km downwind, with scoria fragments, irregular and broken volcanic glass drops (a), some of them elongated (b).

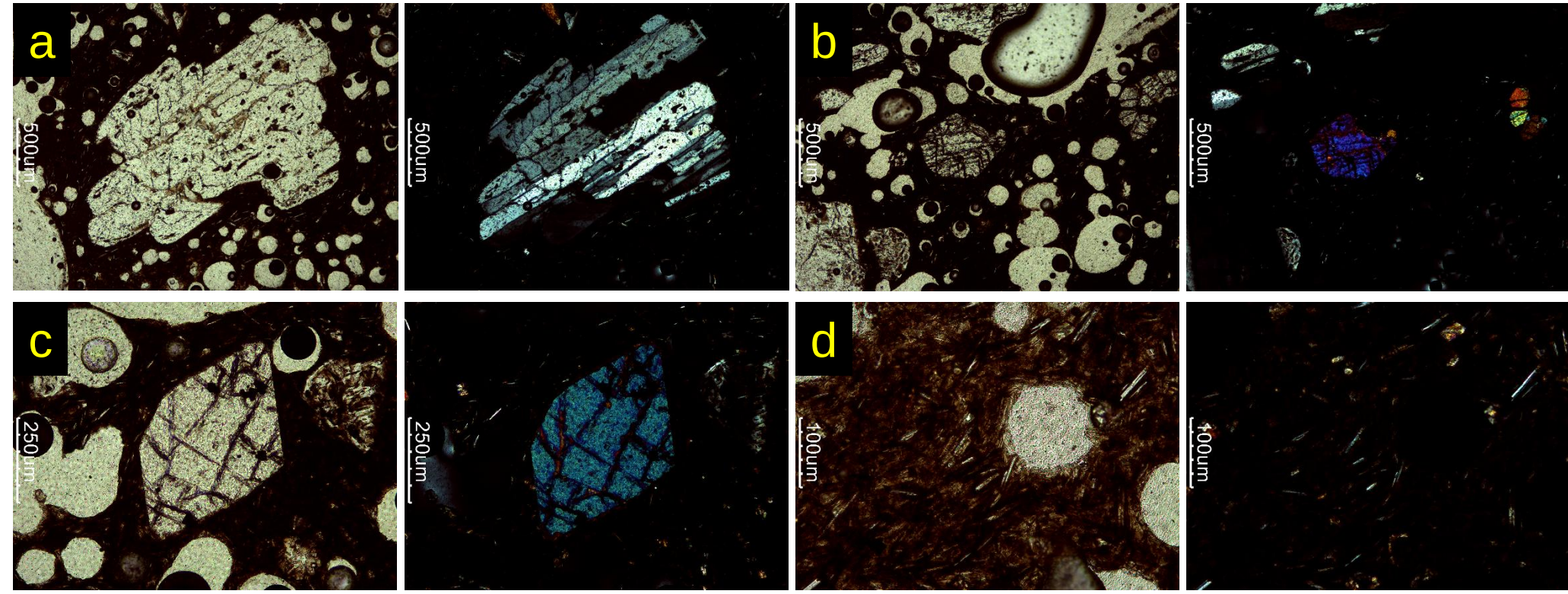


Fig. 5: Mineral assemblage as seen in plane polarized light (left) and crossed nicols (right) of a spatter sample collected 3.7 km northeast from the crater. **a:** Plagioclase phenocrysts. **b:** Olivine phenocrysts. **c:** Clinopyroxene phenocrysts. **d:** microlites.

Magnitude and intensity

With a total duration of 18 minutes for the paroxysmal phase of the eruption, the Mass Eruption Rate (MER) and Volume Discharge Rate (VDR) are obtained from the eruption's total mass and DRE volume, respectively (Table 1). If the [20] magnitude (M) and intensity (I) scales are applied, this eruption was M~-0.38 and I ~6.14. The column height was also calculated from the model of [21] and it ranges from ~10.8 to ~12.3 km a.s.l (Table 1).

Method of isopach Integration	Bulk Volume (km³)	Mass (kg) (ρ=0.76)	DRE Volume(m³)(ρ=2.75)	VDR m³/s	MER kg/s	I	M	Column height, km		VEI
								VDR	MER	
Exp. Thinning	4.95E-03	3.76E+06	1.37E+06	1.27E+03	1.27E+03	6.10	-0.42	10.62	11.10	1
Segment 1	1.66E-03	1.26E+06	4.60E+05	4.26E+02	4.26E+02	5.63	-0.90	8.01	8.37	1
Segment 2	3.29E-03	2.50E+06	9.08E+05	8.41E+02	8.41E+02	5.92	-0.60	9.55	9.98	1
Power Law	4.41E-03	3.35E+06	1.22E+06	1.13E+03	1.13E+03	6.05	-0.48	10.31	10.77	1
Weibull	7.32E-03	5.56E+06	2.02E+06	1.87E+03	1.87E+03	6.27	-0.25	11.76	12.29	1

Table 1: Physical parameters obtained for the tephra fall deposits of the 3 March eruption

IMPACTS OF THE ERUPTION

Impacts include the evacuation process of ca. 3000 people, destruction of roads and private infrastructure at the ski center and "Cavernas Volcánicas" sites, respectively. Water supply remained suspended for >2000 citizens, a week after of the eruption. Impacts from tephra fallout were not reported. Tourism has been severely affected by the closure of the volcano accesses for climbers, and it is the main livelihood activity of Pucón. The rising of technical alerts by the Volcanological Observatory from Southern Andes (Ovdas) during half February prepared the citizens successfully for an eventual eruption even when the red alert was published minutes after the eruption onset.



Fig. 6: Rolling pyroclastic rocks down the flanks of the volcano, revealing the genetic mechanism of the "volcanic mixed avalanches" during the morning of 3 March 2015. Photo by Anselmo Durán.

DISCUSSION AND CONCLUSIONS

The March 2015 eruption shows an interesting chronology of precursory events: 1) the deepening of the lava lake ~1 year before the eruption, 2) the rising of the lava lake on January 2015, 3) the start of the unrest on February 2015 featured by an increase in the seismic and surface activity and 4) a sudden increase in seismicity just hours prior the eruption. If compared, the petrology of the spatter sample is very similar to that of an aerodynamic bomb analyzed by [19]. These fragments share textural and petrological characteristics with the juvenile material reported by [10] for the activity on 2004. In addition, glass composition from 2004 and 2015 products is relatively stable with 54.07 and 56.7 SiO₂ wt%, defining the composition as medium K₂O basaltic andesite to andesite, respectively [10, 19]. Although integrated experiments have contributed to define a system for Villarrica volcano during events with a mild bubble bursting, degassed shallow-system occasionally interrupted by short-lived, rapidly ascending, less-degassed, batches [10] is highly needed to determine the conduit conditions that trigger violent Strombolian activity as observed in 2015, due to the diversity, extent and nature of its volcanic phenomena. This could also contribute to the understanding of depth-fluctuating lava lakes. Finally, the 2015 eruption with M<0 shows characteristics typical of discrete explosions during Hawaiian or Strombolian activity lasting for only seconds to tens of seconds, and releasing very small quantities of material [20].

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